

- 1 (a) 5 (accept 6)  
minimum of 3 points is required to draw a best fit straight line, hence if the maximum point is used for both the increasing and decreasing lines, then a minimum of 5 experiments are required (otherwise 6).

- (b) (i) the total volume is kept constant at 50 cm<sup>3</sup> to ensure a constant heat capacity is used throughout / to ensure that the  $\Delta T$  determined is proportional to each volume of **FA 1** used

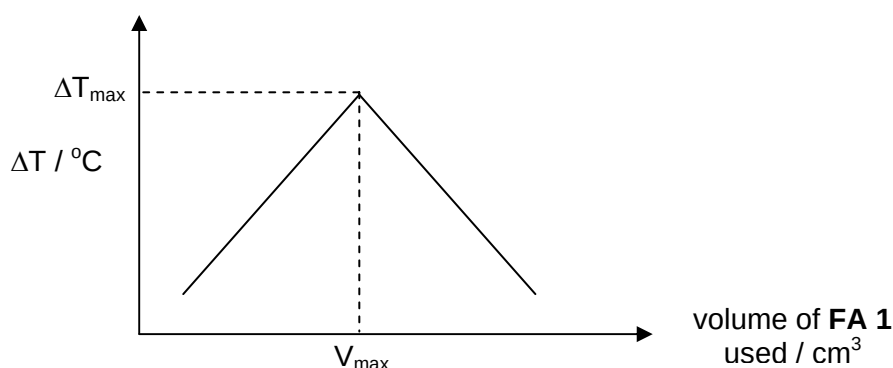
$$\Delta T = - \frac{\Delta H_{\text{neut}} [\text{FA1}]}{mc} (V_{\text{FA1}})$$

- (ii) the volume of FA 1 used must be
- enough to cover the bulb of the thermometer
  - enough to produce significant  $\Delta T$
  - able to minimise percentage error from measuring its volume

- (c) (i) Styrofoam / polystyrene

- (ii) burette

- (d)



volume of FA 1 added which gives maximum  $\Delta T$ ,  $V_{\text{max}}$   
and  
maximum  $\Delta T$ ,  $\Delta T_{\text{max}}$

- (e) (i)  $[\text{FA 1}] = \frac{(50 - V_{\text{max}})(1.00)}{2V_{\text{max}}} = \frac{50 - V_{\text{max}}}{2V_{\text{max}}}$

- (ii) amount of H<sub>2</sub>O formed = amount of NaOH reacted  
 $= [(50 - V_{\text{max}}) \times 10^{-3}](1.00)$   
 $q = (50)(4.2)(\Delta T_{\text{max}}) \text{ J}$

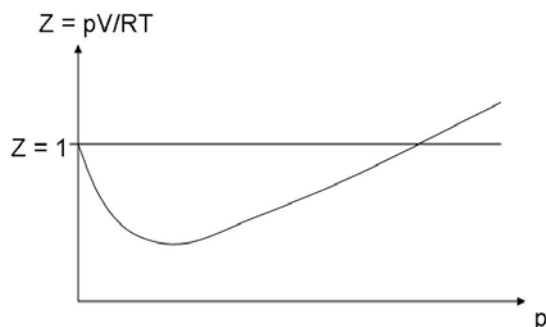
$$\Delta H_{\text{neut}} = - \frac{(50)(4.2)\Delta T}{(50 - V_{\text{max}})} \text{ kJ mol}^{-1}$$

- (f) Step 1: Add x cm<sup>3</sup> of 1<sup>st</sup> unknown into the Styrofoam cup and record its initial temperature.  
 Step 2: Add 2x cm<sup>3</sup> of **FA 3** into another cup and record its initial temperature.  
 Step 3: Transfer the cup with unknown into that containing **FA 3**, stir and record the maximum temperature reached.  
 Step 4: Wash and dry the cups.  
 Step 5: Repeat step 1 to 3, replacing the 1<sup>st</sup> unknown with the 2<sup>nd</sup>.

The unknown that gives a  $\Delta T$  that is **half** the other will be **hydrochloric acid** while the other will be sulfuric acid. As NaOH (**FA 3**) is in excess, the amount of heat from the reactions is dependent on amount of  $H^+$  per unit volume of the acid used. Hence the  $\Delta T$  of the reaction is proportional to the basicity of the acid present in the unknown.

- 2 (a) (i) negligible intermolecular forces of attraction.  
volume of the gas molecules is negligible compared to the volume it occupies.

(ii)



(iii) sp

- (b) (i) When gas is compressed, the molecules are very close together, and the intermolecular forces of attraction become significant.
- (ii) expansion into low pressure requires the molecules to overcome intermolecular forces of attraction in order to move apart.  
hence, heat is absorbed by the molecules from the surroundings, making the gas cool.
- (c) (i) Ammonia gas has higher critical temperature than carbon dioxide. Stronger intermolecular forces of attraction (hydrogen bonding compared to van der Waal's), requires higher temperature to prevent gas from being liquefied.
- (ii) The fluorinated alkyl group in the surfactant can dissolve well in the carbon dioxide solvent (due to  $\pi$ - $\pi$  between polar ends of  $CO_2$  and C-F bond) while its hydrocarbon chain can interact favourably with oils (id-id).

*Thus the surfactant molecules will help to bind to oil from clothes and is removed together by carbon dioxide, making it a good dry cleaning solvent.*

- 3 (a) (i)  $M_r$  of  $PCl_5$  = **208.5**  
 $M_r$  of  $PCl_3$  = **137.5**  
 $M_r$  of  $Cl_2$  = **71.0**

- (ii) average molar mass of the gaseous mixture = 
$$\frac{(15.0)(8.31)(473)}{(3.10 \times 10^5)(1 \times 10^{-3})}$$
  
$$= 190 \text{ g mol}^{-1}$$

average  $M_r$  of the gaseous mixture = **190**

$$\text{(iii) fraction dissociated (x)} = \frac{208.5 - 190}{190} = 0.0963$$

$$K_c = \frac{(0.0963^2)}{1 - 0.0963} = 0.0103 = 1.0 \times 10^{-2} \text{ (shown)}$$

$$\text{(iv) } \Delta G^\ominus = -2.303(8.31)(473) \lg (1.0 \times 10^{-2}) = +18104 \text{ J mol}^{-1} = +18.1 \text{ kJ mol}^{-1}$$

$$\text{(v) } \Delta H = \Delta H_f(\text{PCl}_3) - \Delta H_f(\text{PCl}_5) = -319 \text{ kJ mol}^{-1} - (-443 \text{ kJ mol}^{-1}) = +124 \text{ kJ mol}^{-1}$$

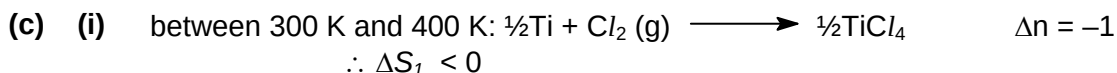
$$\text{(vi) } \Delta G = \Delta H - T\Delta S$$

$$\Delta S = \frac{124 - (18.1)}{473} = +0.224 \text{ kJ mol}^{-1} \text{ K}^{-1} \text{ (OR } +224 \text{ J mol}^{-1} \text{ K}^{-1})$$

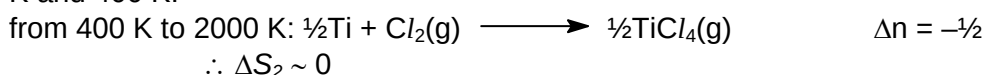
(b) for extraction to be feasible,  $\Delta G < 0$

$$T > \frac{720}{0.365} > 1970 \text{ K}$$

The high temperature required (1970 K) is hard to maintain / high operating cost to maintain high temperature.



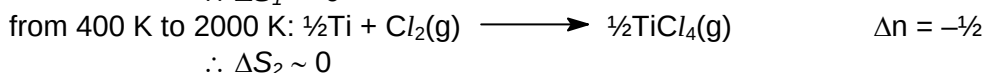
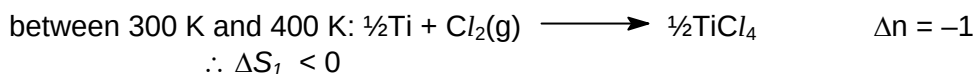
Since  $\Delta H$  and  $\Delta S$  are both negative,  $-T\Delta S$  becomes more positive with increasing temperature and hence the graph for reaction 1 becomes more positive between 300 K and 400 K.



Since  $\Delta S$  is small,  $-T\Delta S$  remains relatively constant with increasing temperature and hence the graph for reaction 1 becomes almost independent of temperature from 400 K to 2000 K.

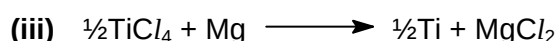
OR

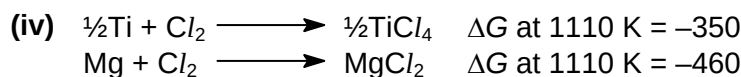
for graph of  $\Delta G$  against  $T$ , y-intercept gives the  $\Delta H$  and the gradient gives  $-\Delta S$



hence the graph shows an increasing line from 300 K to 400 K and a horizontal line (gradient = 0) from 400 K to 2000 K.

(ii) To prevent the hot titanium extracted from oxidising back to  $\text{TiO}_2$ .  
*(fyi: traces of oxygen or nitrogen in the titanium tend to make the metal brittle)*

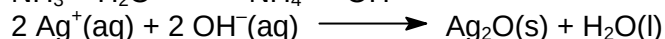
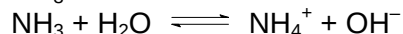




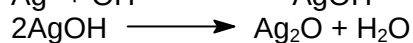
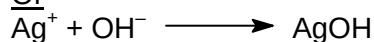
$$\Delta G = -460 - (-350) \\ = -110 \text{ kJ mol}^{-1} \quad [\text{depend on balanced equation in (c)(iii)}]$$

(v) **1870 K**  
 (at T higher than 1870 K,  $\Delta G > 0$ )

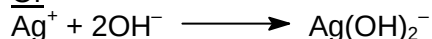
4 (a) (i)  $\text{NH}_3$  is a weak base which dissociates in water to give hydroxide ions.



Or



Or



(ii)  $[\text{Ag}(\text{NH}_3)_2]^+$

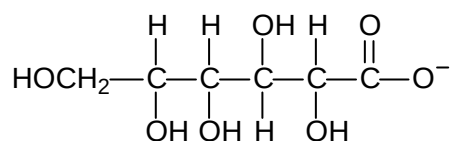
(iii)  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10}$

(iv) fully-filled d orbitals (absence of empty or partially filled d orbitals)  
no d-d transition can occur.

*for d electrons to be promoted to higher energy level, it has to absorb energy outside visible light spectrum and hence appear colourless*

(v) mirror can be applied onto the glassware by immersing it in solution containing Tollen's reagent and glucose solution (or suitable reducing group e.g. aldehyde) and warmed.

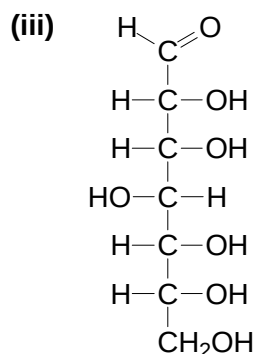
(b) (i) Glucose, brick red ppt of  $\text{Cu}_2\text{O}$ .



(ii) Step 1: HCN, NaCN catalyst  
 Step 2: dilute  $\text{H}_2\text{SO}_4$ , heat under reflux

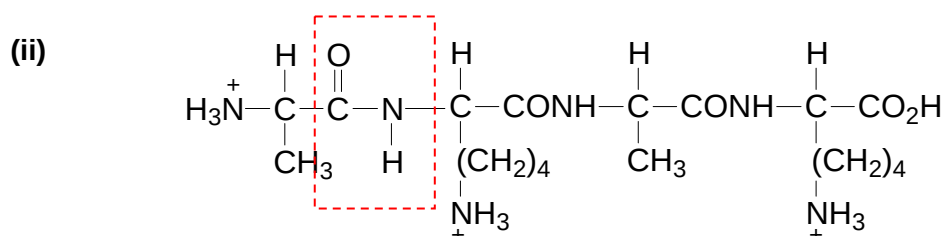
Intermediate Y:  $\text{RCH}(\text{OH})\text{CH}(\text{OH})\text{CN}$

Intermediate Z:  $\text{RCH}(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{OH}$



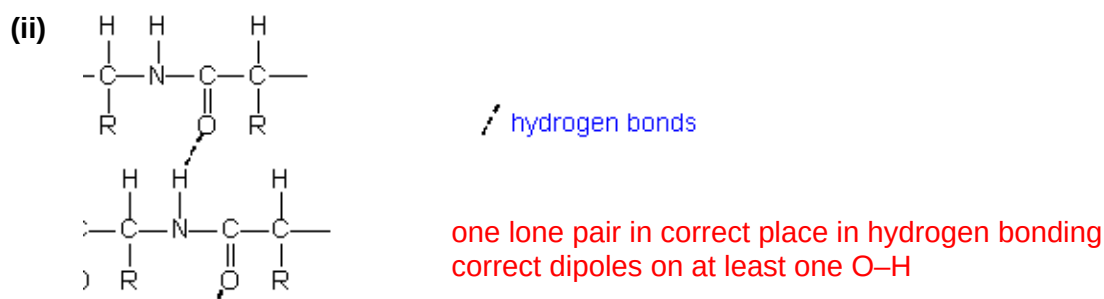
(iv) Type of reaction: nucleophilic addition

- 5 (a) (i) The polypeptides contains **9 amino residues** as it has 8 peptide bonds  
(in words or in drawing)  
By observation the order is ala-ala-ser-asp-ala-ala-lys-gly-lys



must show one –CONH– peptide group in fully displayed form

- (b) (i) Denaturation of proteins involves the disruption of the secondary, tertiary and quaternary structures as these are held together by relatively weak attractions.



- (iii)
- van der Waals' forces exist between non-polar R groups such as –CH<sub>3</sub> of alanine
  - electrostatic forces of attraction exist between charged R groups such as –CH<sub>2</sub>CO<sub>2</sub><sup>–</sup> group of anionic aspartic acid and the –(CH<sub>2</sub>)<sub>4</sub>NH<sub>3</sub><sup>+</sup> group of the cationic lysine
  - hydrogen bonding exists between polar R groups such as the –CH<sub>2</sub>OH group of serine can form hydrogen bonds with the amine group in –CH<sub>2</sub>NH<sub>2</sub> group of lysine

- (c) (i) Addition of a little OH<sup>–</sup>  

$$^+\text{H}_3\text{NCH}(\text{CH}_3)\text{CO}_2^- + \text{OH}^- \longrightarrow \text{H}_2\text{NCH}(\text{CH}_3)\text{CO}_2^- + \text{H}_2\text{O}$$

(ii)

|                                                                    |                                                              |
|--------------------------------------------------------------------|--------------------------------------------------------------|
| <b>J:</b> $^+\text{H}_3\text{NCH}(\text{CH}_3)\text{CO}_2\text{H}$ | <b>K:</b> $^+\text{H}_3\text{NCH}(\text{CH}_3)\text{CO}_2^-$ |
| <b>L:</b> $\text{H}_2\text{NCH}(\text{CH}_3)\text{CO}_2^-$         |                                                              |

- (iii) pH 6